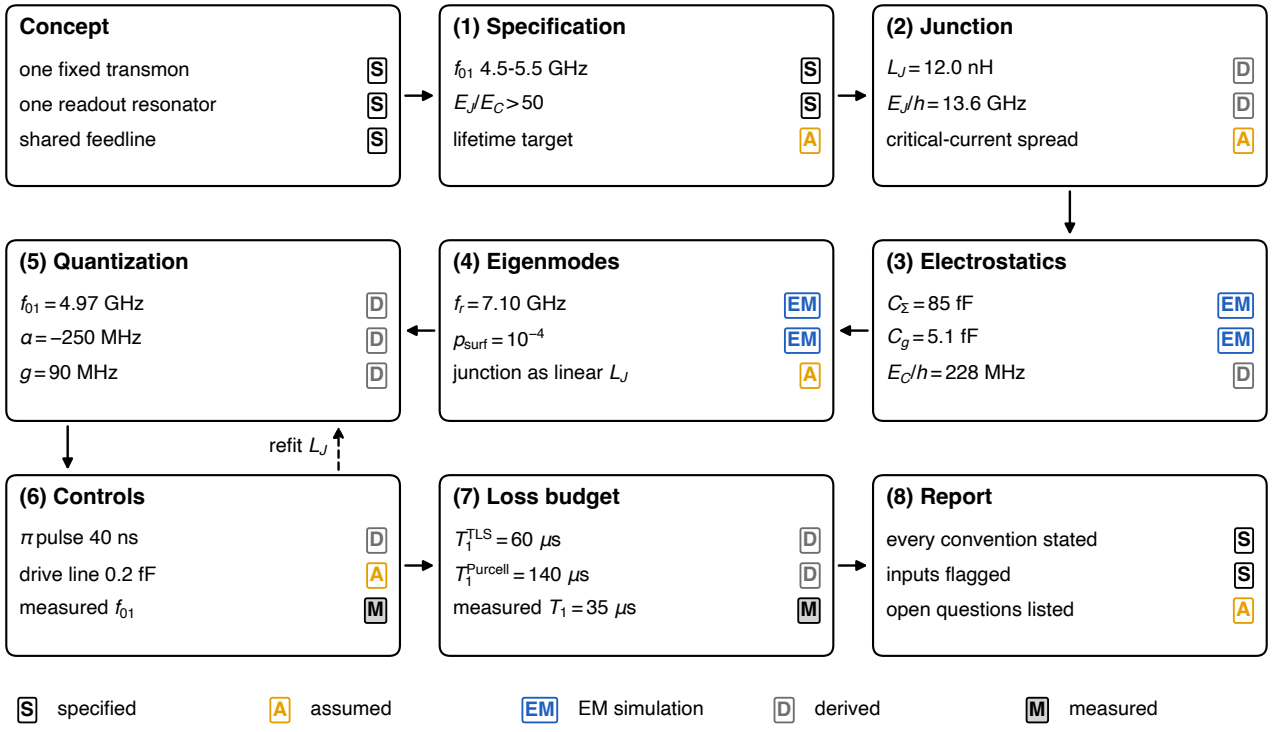
Concept
one fixed transmon S
one readout resonator S
shared feedline S
(1) Specification
f_{01} 4.5-5.5 GHz S
$E_J/E_C > 50$ S
lifetime target A
(2) Junction
$L_J = 12.0$ nH D
$E_J/h = 13.6$ GHz D
critical-current spread A
(3) Electrostatics
$C_\Sigma = 85$ fF EM
$C_g = 5.1$ fF EM
$E_C/h = 228$ MHz D
(4) Eigenmodes
$f_r = 7.10$ GHz EM
$p_{surf} = 10^{-4}$ EM
junction as linear L_J A
(5) Quantization
$f_{01} = 4.97$ GHz D
$\alpha = -250$ MHz D
$g = 90$ MHz D
refit L_J
(6) Controls
π pulse 40 ns D
drive line 0.2 fF A
measured f_{01} M
(7) Loss budget
$T_1^{TLS} = 60$ μs D
$T_1^{Purcell} = 140$ μs D
measured $T_1 = 35$ μs M
(8) Report
every convention stated S
inputs flagged S
open questions listed A
S specified
A assumed
EM EM simulation
D derived
M measured